



Sunbeam Institute of Information Technology

Pune and Karad

Module – Internet of Things (IoT)

Trainer - Devendra Dhande

Email – devendra.dhande@sunbeaminfo.com

- It is about taking suitable action to maximize reward in a particular situation
- It is employed by various software and machines to find the best possible behavior or path it should take in a specific situation
- Reinforcement learning differs from the supervised learning in a way that in supervised learning the training data has the answer key with it so the model is trained with the correct answer itself whereas in reinforcement learning, there is no answer but the reinforcement agent decides what to do to perform the given task
- In the absence of training dataset, it is bound to learn from its experience
- Examples
 - Resources management in computer clusters
 - Traffic Light Control
 - Robotics
 - Web system configuration
 - Chemistry
- Algorithms
 - Q-Learning
 - Deep Q-Learning

- Look at the big picture
- Get the data
- Discover and visualize the data to gain insights
- Prepare the data for Machine Learning algorithms
- Select a model and train it
- Fine-tune your model
- Present your solution
- Launch, monitor, and maintain your system

- Frame the Problem
 - The first question to ask your boss is what exactly the business objective is
 - Building a model is probably not the end goal
 - How does the company expect to use and benefit from this model?
 - Knowing the objective is important because it will determine
 - how you frame the problem
 - which algorithms you will select
 - which performance measure you will use to evaluate your model
 - how much effort you will spend tweaking it
- Select a Performance Measure
 - Your next step is to select a performance measure
 - A typical performance measure for regression problems is the Root Mean Square Error (RMSE)
 - It gives an idea of how much error the system typically makes in its predictions, with a higher weight for large errors

Get the data

- Decide the data source
- Download the data and make it available for the further learning
- Take a Quick Look at the Data Structure
 - Understand the data set and features-m
 - Evaluate the features and decide which one(s) are needed
- Create a Test Set
 - Keep some records aside for testing and validation



Discover and Visualize the Data to Gain Insights

- Visualize the data
 - Use libraries like matplotlib or seaborn
 - Understand the pattern and relationship
- Look for correlation
- Experiment with attribute combinations



Prepare the Data for Machine Learning Algorithms

- Data Cleaning
 - Process of cleaning the data set to prepare it for ML algorithm
 - Steps
 - Check for the missing data
 - Check for wrong data types
 - Add features if needed
 - Remove unwanted features
- Feature Scaling
 - ML algorithms don't perform well when the input numerical attributes have very different scale
 - Scale the features to bring all of them to a single scale
- Handle categorical / text data
 - Use transformers to convert categorical to numerical

Select and Train a Model

- Training the model using train data set
 - Create a model using selected algorithm
 - Save the model for future use
- Evaluation the model
 - Evaluate the model to see if there is any chance to improve the accuracy
 - Techniques
 - Cross Validation

- **Grid Search**
 - One option would be to fiddle with the hyperparameters manually, until you find a great combination of hyperparameter values
 - This would be very tedious work, and you may not have time to explore many combinations
 - You can also automate this process using libraries like sci-kit
- **Randomized Search**
 - The grid search approach is fine when you are exploring relatively few combinations
 - But when the hyperparameter search space is large, it is often preferable to use randomized search
- **Ensemble Methods**
 - Another way to fine-tune your system is to try to combine the models that perform best
 - The group (or “ensemble”) will often perform better than the best individual model, especially if the individual models make very different types of errors.
- Analyze the Best Models and Their Errors
- Evaluate Your System on the Test Set



Launch, Monitor, and Maintain Your System

- Deploy the application for the end users
- Monitor the application's performance
- If the data keeps evolving, update your datasets and retrain your model regularly
- You should probably automate the whole process as much as possible
 - Collect fresh data regularly and label it
 - Write a script to train the model and fine-tune the hyperparameters automatically. This script could run automatically, for example every day or every week, depending on your needs
 - Write another script that will evaluate both the new model and the previous model on the updated test set, and deploy the model to production if the performance has not decreased (if it did, make sure you investigate why)





- Feature Engineering is the process of creating new features or transforming existing features to improve the performance of a machine-learning model.
- Process of transforming raw data into features that are suitable for machine learning models
- Involve techniques that enable to create new features by combining or transforming the existing ones
- A feature (also known as a variable or attribute) is an individual measurable property or characteristic of a data point that is used as input for a machine learning algorithm.
- Features can be numerical, categorical, or text-based, and they represent different aspects of the data
- Feature engineering in Machine learning consists of mainly 5 processes:
 - Feature Creation
 - Feature Transformation
 - Feature Extraction
 - Feature Selection
 - Feature Scaling.

- Feature Creation is the process of generating new features based on domain knowledge or by observing patterns in the data.
- **Types of Feature Creation:**
 - **Domain-Specific:**
 - Creating new features based on domain knowledge, such as creating features based on business rules or industry standards.
 - **Data-Driven:**
 - Creating new features by observing patterns in the data, such as calculating aggregations or creating interaction features.
 - **Synthetic:**
 - Generating new features by combining existing features or synthesizing new data points.

- Feature Transformation is the process of transforming the features into a more suitable representation for the machine learning model.
- **Types of Feature Transformation:**
 - **Normalization:**
 - Rescaling the features to have a similar range, such as between 0 and 1, to prevent some features from dominating others.
 - **Scaling:**
 - Scaling is a technique used to transform numerical variables to have a similar scale, so that they can be compared more easily. Rescaling the features to have a similar scale, such as having a standard deviation of 1, to make sure the model considers all features equally.
 - **Encoding:**
 - Transforming categorical features into a numerical representation. Examples are one-hot encoding and label encoding.
 - **Transformation:**
 - Transforming the features using mathematical operations to change the distribution or scale of the features. Examples are logarithmic, square root, and reciprocal transformations.

- Feature Extraction is the process of creating new features from existing ones to provide more relevant information to the machine learning model.
- **Types of Feature Extraction:**
 - **Dimensionality Reduction:**
 - Reducing the number of features by transforming the data into a lower-dimensional space while retaining important information.
 - **Feature Combination:**
 - Combining two or more existing features to create a new one. For example, the interaction between two features.
 - **Feature Aggregation:**
 - Aggregating features to create a new one. For example, calculating the mean, sum, or count of a set of features.
 - **Feature Transformation:**
 - Transforming existing features into a new representation. For example, log transformation of a feature with a skewed distribution.

- Feature Selection is the process of selecting a subset of relevant features from the dataset to be used in a machine-learning model.
- **Types of Feature Selection:**
 - **Filter Method:**
 - Based on the statistical measure of the relationship between the feature and the target variable. Features with a high correlation are selected.
 - **Wrapper Method:**
 - Based on the evaluation of the feature subset using a specific machine learning algorithm. The feature subset that results in the best performance is selected.
 - **Embedded Method:**
 - Based on the feature selection as part of the training process of the machine learning algorithm.

- Feature Scaling is the process of transforming the features so that they have a similar scale.
- **Types of Feature Scaling:**
 - **Min-Max Scaling:**
 - Rescaling the features to a specific range, such as between 0 and 1, by subtracting the minimum value and dividing by the range.
 - **Standard Scaling:**
 - Rescaling the features to have a mean of 0 and a standard deviation of 1 by subtracting the mean and dividing by the standard deviation.
 - **Robust Scaling:**
 - Rescaling the features to be robust to outliers by dividing them by the interquartile range.

Variable / Column / Feature

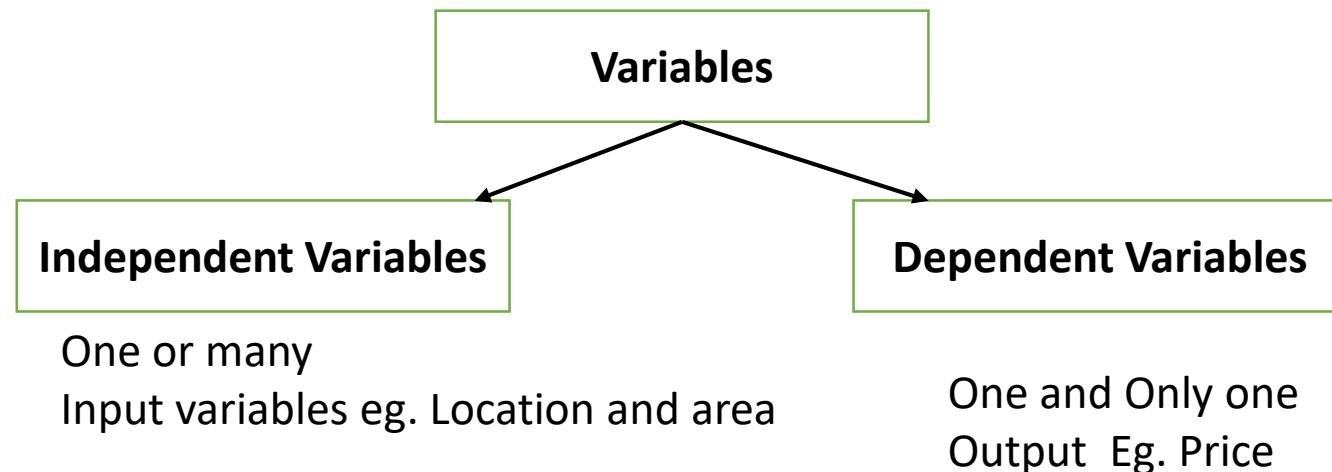
No.	Location	Area	Price
1	Pune	Abc	70
2	Mumbai	Pqr	90

**Known Data
Or
Labelled Data**

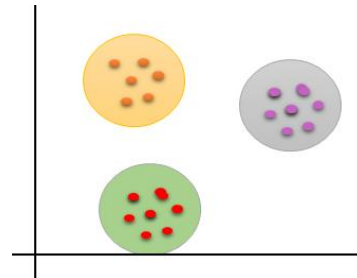
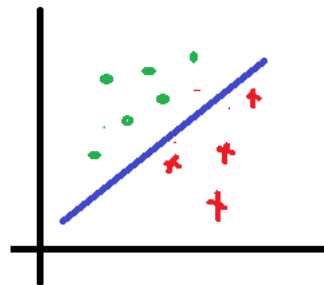
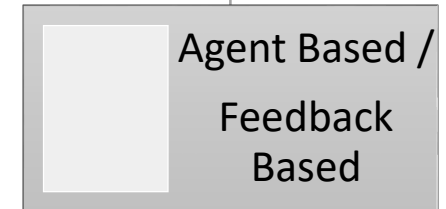
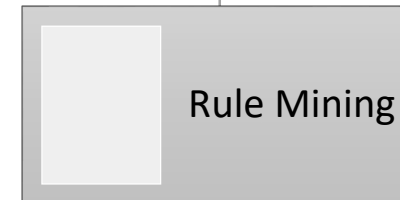
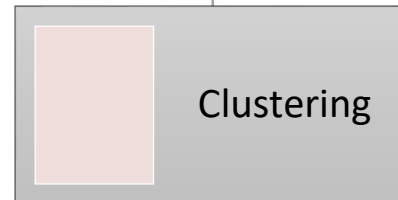
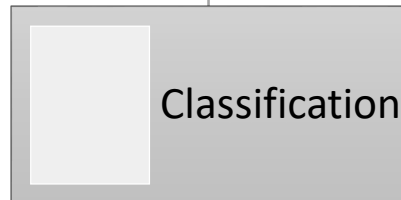
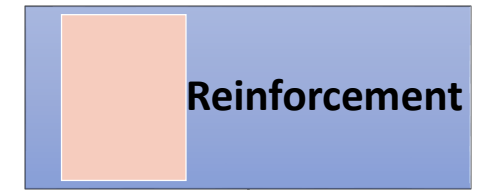
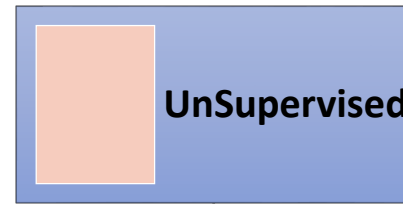
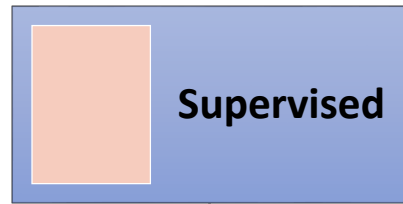
No. column not required

Location and Area are Independent variables

Price is Dependent Variable



Machine Learning



Association
Product1 -> product 2
Bread -> butter

Binary = 2 classes / labels
Multiclass means >2



Performance Measures

Regression

- RMSE (Root Mean Squared Error) :square root of Mean Squared error
- MAE (Mean Absolute Error) : the average of the absolute difference between the actual and predicted values in the dataset
- MSE :the average of the squared difference between the original and predicted values in the data set

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y})^2}$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}|$$

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y})^2$$

Where,

$\hat{y}$ – predicted value of y
 $\bar{y}$ – mean value of y

The lower value of MAE, MSE, and RMSE implies higher accuracy of a regression model



Performance Measures

Classification

- Accuracy Score : measures how often the classifier correctly predicts
- F1-Score : the harmonic mean of precision and recall.
- Confusion Matrix : It is a matrix of size 2×2 for binary classification with actual values on one axis and predicted on another

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$F1 = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$\text{Precision} = \frac{\text{TruePositive}}{\text{TruePositive} + \text{FalsePositive}}$$

$$\text{Recall} = \frac{\text{TruePositive}}{\text{TruePositive} + \text{FalseNegative}}$$

		ACTUAL	
		Negative	Positive
PREDICTION	Negative	TRUE NEGATIVE	FALSE NEGATIVE
	Positive	FALSE POSITIVE	TRUE POSITIVE



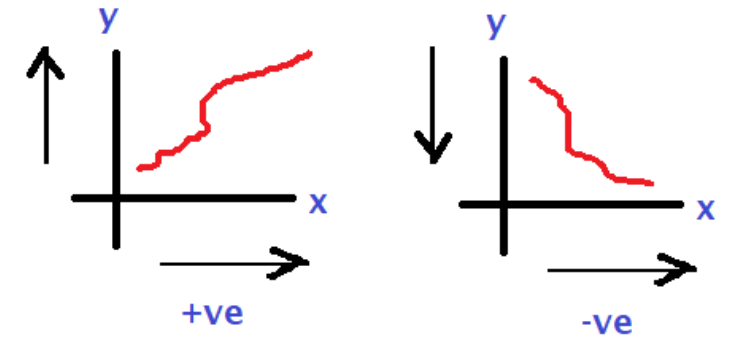
Terminology

■ Univariate

- This type of data consists of only one variable (eg. Temperature value)
- It does not deal with causes or relationships
- the main purpose of the analysis is to describe the data and find patterns that exist within it

■ Bivariate

- This type of data involves two different variables
- The analysis of this type of data deals with causes and relationships
- the analysis is done to find out the relationship among the two variables



■ Multivariate

- When the data involves three or more variables
- It is similar to bivariate but contains more than one independent variable



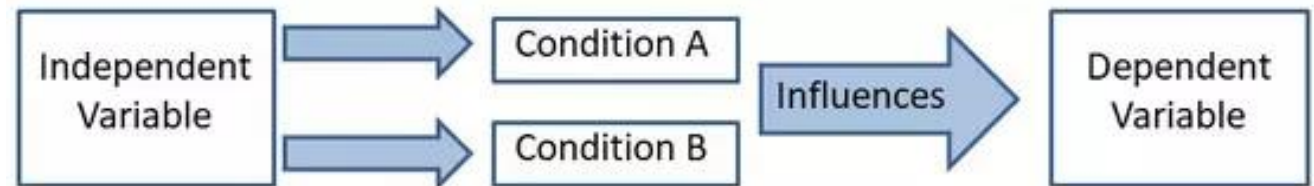
Terminology

- **Independent variable(s)**

- A variable that represents a quantity that is being manipulated in an experiment
- **Represents input**
- Also known as regressors in a statistical context.
- x is often the variable used to represent the independent variable in an equation

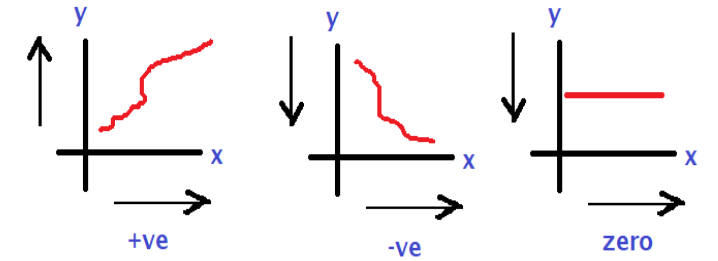
- **Dependent variable**

- A quantity whose value *depends* on how the independent variable is manipulated
- **Represents output**
- y is often the variable used to represent the independent variable in an equation



Covariance

- A measure of the relationship between two random variables
- The metric evaluates how much – to what extent – the variables change together
- A positive covariance would indicate a positive linear relationship between the variables
- A negative covariance would indicate the opposite



$$cov(x, y) = \frac{\sum (X_i - \mu)(Y_i - \mu)}{N}$$

Population

$$cov(x, y) = \frac{\sum (X_i - \bar{x})(Y_i - \bar{y})}{n}$$

Sample

- Where
 - X_i – the values of the X-variable
 - Y_i – the values of the Y-variable
 - $\bar{x}$ – the mean (average) of the X-variable
 - $\bar{y}$ – the mean (average) of the Y-variable
 - n – the number of the data points in sample
 - N – the number of the data points in Population

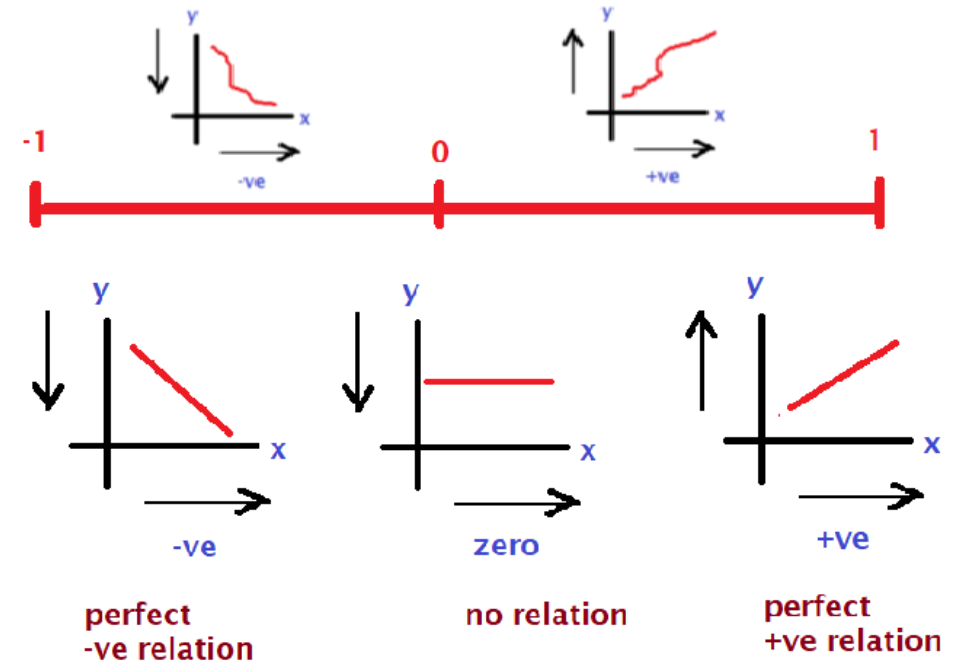


Correlation

- Measures the strength of the relationship between variables
- Correlation is the scaled measure of covariance
- It is dimensionless: the correlation coefficient is always a pure value and not measured in any units

$$\rho(X,Y) = \frac{cov(X,Y)}{\sigma_x\sigma_y}$$

- Where
 - $\rho(X,Y)$ – the correlation between the variables X and Y
 - $cov(X,Y)$ – the covariance between the variables X and Y
 - σ_x – the standard deviation of the X-variable
 - σ_y – the standard deviation of the Y-variable



Correlation coefficient

- Following are the ways to calculate correlation coefficient
 - Karl Pearson's coefficient of correlation
 - Spearman's Rank correlation
 - Scatter diagram
 - Coefficient of concurrent duration
- Correlation (r)
 - $-1 \leq r \leq 1$
 - $r = 1$ (perfect correlation)
 - $r = -1$ (perfect negative correlation)
 - $r > 0$ (positive correlation)
 - $r < 0$ (negative correlation)
 - $r = 0$ (no correlation)

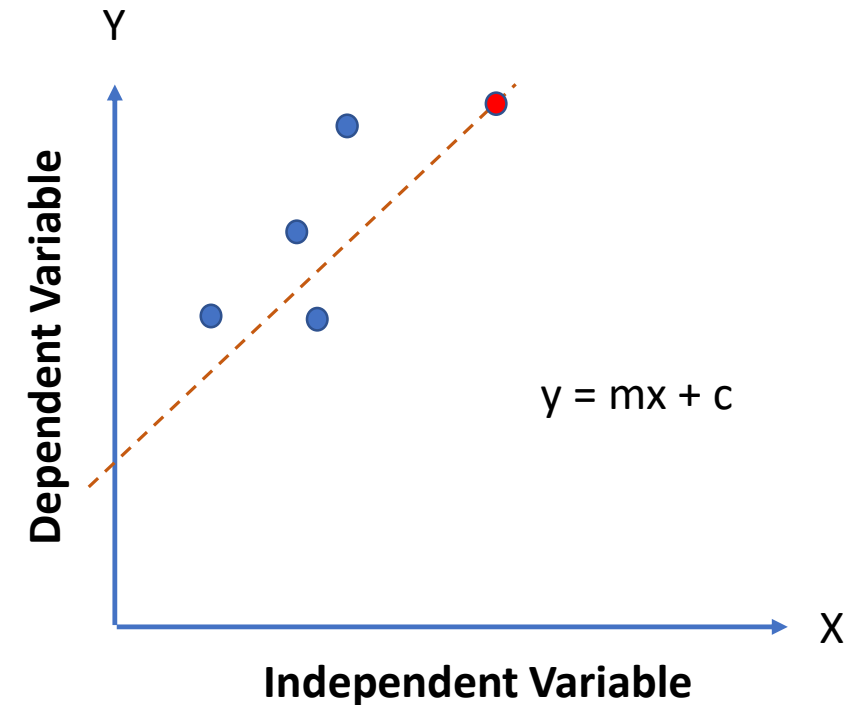


Linear Regression



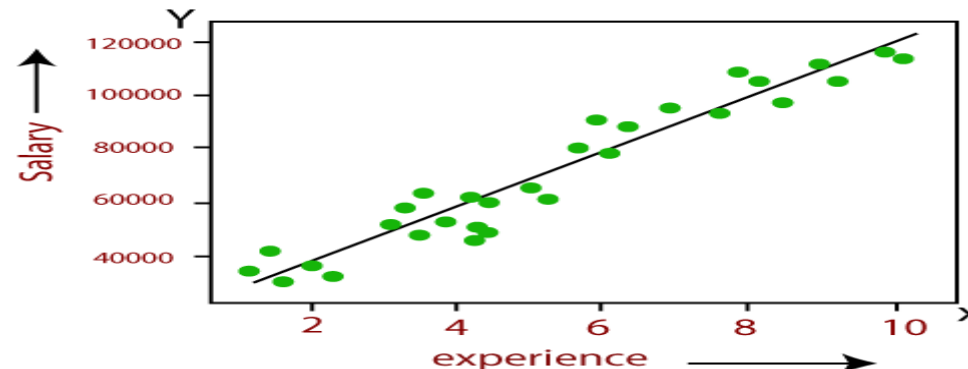
Overview

- The data in Linear Regression is modelled using a straight line
- Linear regression is a statistical regression method which is used for predictive analysis.
- It is one of the very simple and easy algorithms which works on regression and shows the relationship between the continuous variables.
- It is used with continuous variable
- It gives a future value as an output
- To calculate accuracy following methods are used
 - R-squared
 - Adjusted R-squared



Simple and Multiple Linear Regression

- If there is only one input variable (x), then such linear regression is called **simple linear regression**.
- If there is more than one input variable, then such linear regression is called **multiple linear regression**.
- Predicting the salary of an employee on the basis of **the year of experience**.
- $Y = aX + b$
- Y = dependent variables (target variables),
X = Independent variables (predictor variables),
a and b are the linear coefficients
- Eg . Analyzing trends and sales estimates
- Eg. Salary forecasting



Regression Equation

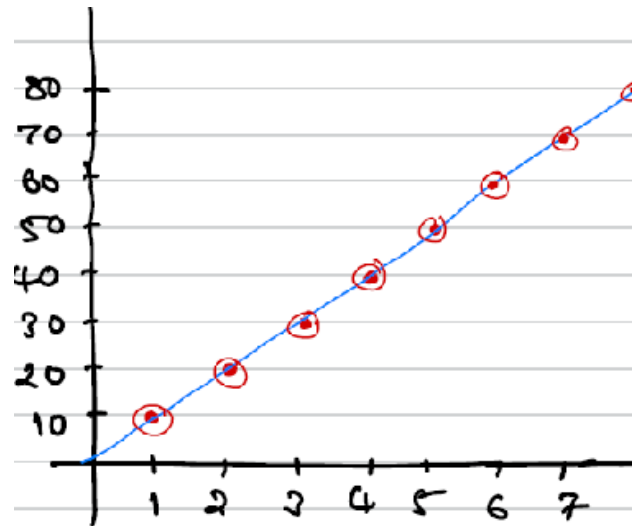
X	Y
1	10
2	20
3	30
4	40
5	?

$$\bar{Y} = 25$$

X	Y	$Y - \bar{Y}$	$(Y - \bar{Y})^2$
1	10	-15	225
2	20	-5	25
3	30	5	25
4	40	15	225

What likely to be the value of Y if X = 5

$$\begin{aligned} Y &= 10 * x \\ &= 10 * 5 \\ &= 50 \end{aligned}$$



Regression Model Evaluation Metrics

- For evaluation of regression model, following metrics are used
 - MAE
 - MSE
 - RMSE
 - R2
 - Adjusted R2



Mean Absolute Error (MAE)

- The MAE measures the average magnitude of the errors in a set of forecasts, without considering their direction (absolute)
- It measures accuracy for continuous variables
- The MAE is the average over the verification sample of the absolute values of the differences between forecast and the corresponding observation $|y - \hat{y}|$
- The MAE is a linear score which means that all the individual differences are weighted equally in the average

$$\frac{\sum |y - \hat{y}|}{n}$$



Mean Squared Error (MSE)

- In statistics, the mean squared error (MSE) or mean squared deviation (MSD) of an estimator (of a procedure for estimating an unobserved quantity) measures the average of the squares of the error
- That is, the average squared difference between the estimated values and the actual value
- MSE is a risk function, corresponding to the expected value of the squared error loss
- The fact that MSE is almost always strictly positive (and not zero) is because of randomness or because the estimator does not account for information that could produce a more accurate estimate
- The MSE is a measure of the quality of an estimator
- As it is derived from the square of Euclidean distance, it is always a positive value with the error decreasing as the error approaches zero

$$\frac{\sum (y - \hat{y})^2}{n}$$



Root Mean Squared Error (RMSE)

- RMSE is the most popular evaluation metric used in regression problems
- It follows an assumption that error are unbiased and follow a normal distribution
- Here are the key points to consider on RMSE:
 - The power of 'square root' empowers this metric to show large number deviations
 - The 'squared' nature of this metric helps to deliver more robust results which prevents cancelling the positive and negative error values
- It avoids the use of absolute error values which is highly undesirable in mathematical calculations
- When we have more samples, reconstructing the error distribution using RMSE is considered to be more reliable
- RMSE is highly affected by outlier values. Hence, make sure you've removed outliers from your data set prior to using this metric.
- As compared to mean absolute error, RMSE gives higher weightage and punishes large errors



R-Squared (R^2)

- We learned that when the RMSE decreases, the model's performance will improve
- But these values alone are not intuitive
- When we talk about the RMSE metrics, we do not have a benchmark to compare
- This is where we can use R-Squared metric
- In other words how good our regression model as compared to a very simple model that just predicts the mean value of target from the train set as predictions



Adjusted R-Squared

- A model performing equal to baseline would give R-Squared as 0
- Better the model, higher the r2 value
- The best model with all correct predictions would give R-Squared as 1
- However, on adding new features to the model, the R-Squared value either increases or remains the same
- R-Squared does not penalize for adding features that add no value to the model
- So an improved version over the R-Squared is the adjusted R-Squared

$$\bar{R}^2 = 1 - (1 - R^2) \left[\frac{n-1}{n-(k+1)} \right]$$

- k: number of features
- n: number of samples



Classification Model Evaluation Metrics

- For evaluation of classification model, following metrics are used
 - Confusion Matrix
 - F1 Score
 - Auc-Roc



Confusion Matrix

- A confusion matrix is an $N \times N$ matrix, where N is the number of classes being predicted
- The confusion matrix provides more insight into not only the performance of a predictive model, but also which classes are being predicted correctly, which incorrectly, and what type of errors are being made

Observed \ Predicted			Predicted condition		
Observed	Predicted		Total population = $P + N$	Predicted condition positive (PP)	Predicted condition negative (PN)
1	1	TP	Actual condition	Actual condition positive (P)	True positive (TP), hit
1	0	FN			
0	0	TN		Actual condition negative (N)	False positive (FP), Type I error, false alarm, overestimation
0	1	FP			
				False negative (FN), Type II error, miss, underestimation	True negative (TN), correct rejection



TP vs FP vs TN vs FN



Cat



Cat



Cat



Cat



No Cat



No Cat



No Cat



Cat



Cat

TP vs FP vs TN vs FN



FP

Cat



TP

Cat



FP

Cat



TP

Cat



TN

No Cat



FN

No Cat



TN

No Cat



TP

Cat



TP

Cat

TP = 4

FP = 2

Total P = 6

TN = 2

FN = 1

Total N = 3

Total = 9

Accuracy



Cat



Cat



Cat



Cat



No Cat



No Cat



No Cat



Cat



Cat

How many we got right ?

Accuracy : How many we got right?



FP

Cat



TP

Cat



FP

Cat



TP

Cat



TN

No Cat



FN

No Cat



TN

No Cat



TP

Cat



TP

Cat



$$\begin{aligned}\text{Correct} &= \text{TP} + \text{TN} / \text{Total} \\ &= 6 / 9 \\ &= 2/3 \\ &= 0.66\end{aligned}$$

Precision

- Precision talks about how precise/accurate your model is out of those predicted positive, how many of them are actual positive
- Precision is a good measure to determine, when the costs of False Positive is high
- For instance, in email spam detection, a false positive means that an email that is non-spam (actual negative) has been identified as spam (predicted spam). The email user might lose important emails if the precision is not high for the spam detection model.

$$\begin{aligned}\text{Precision} &= \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}} \\ &= \frac{\text{True Positive}}{\text{Total Predicted Positive}}\end{aligned}$$



Precision



Cat



Cat



Cat



Cat



No Cat



No Cat



No Cat



Cat



Cat

Out of all Cat predictions how many we got right ?

Precision : Out of all Cat predictions how many we got right ?



FP

Cat



TP

Cat



FP

Cat



TP

Cat



TN

No Cat



FN

No Cat



TN

No Cat



TP

Cat



TP

Cat

True positive = 4

Total positive = 6

Precision of + ve = $4/6 = 2/3 = 0.66$

True Negative = 2

Total Negative = 3

Precision of -ve = $2/3 = 0.66$

Recall

- Recall actually calculates how many of the Actual Positives our model capture through labelling it as Positive (True Positive)
- Applying the same understanding, we know that Recall shall be the model metric we use to select our best model when there is a high cost associated with False Negative
- For instance, in fraud detection or sick patient detection, if a fraudulent transaction (Actual Positive) is predicted as non-fraudulent (Predicted Negative), the consequence can be very bad for the bank
- Similarly, in sick patient detection, if a sick patient (Actual Positive) goes through the test and predicted as not sick (Predicted Negative), the cost associated with False Negative will be extremely high if the sickness is contagious

$$\begin{aligned}\text{Recall} &= \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}} \\ &= \frac{\text{True Positive}}{\text{Total Actual Positive}}\end{aligned}$$



Recall



Cat



Cat



Cat



Cat



No Cat



No Cat



No Cat



Cat



Cat

Out of all Cat truth how many we got right ?

Recall : Out of all Cat truth how many we got right ?



FP

Cat



TP

Cat



FP

Cat



TP

Cat



TN

No Cat



FN

No Cat



TN

No Cat



TP

Cat



TP

Cat

True positive = 4

Total Actual positive = 5

Recall of + ve = $4/5 = 0.80$

True Negative = 2

Total Actual Negative = 4

Recall of -ve = $2/4 = 0.50$

F1 Score

- The F1 score is the harmonic mean of the precision and recall
- The highest possible value of an F-score is 1.0, indicating perfect precision and recall, and the lowest possible value is 0, if either the precision or the recall is zero
- The F1 score is also known as the Sørensen–Dice coefficient or Dice similarity coefficient (DSC)

$$F1 = 2 \times \frac{Precision * Recall}{Precision + Recall}$$



K-Nearest Neighbours



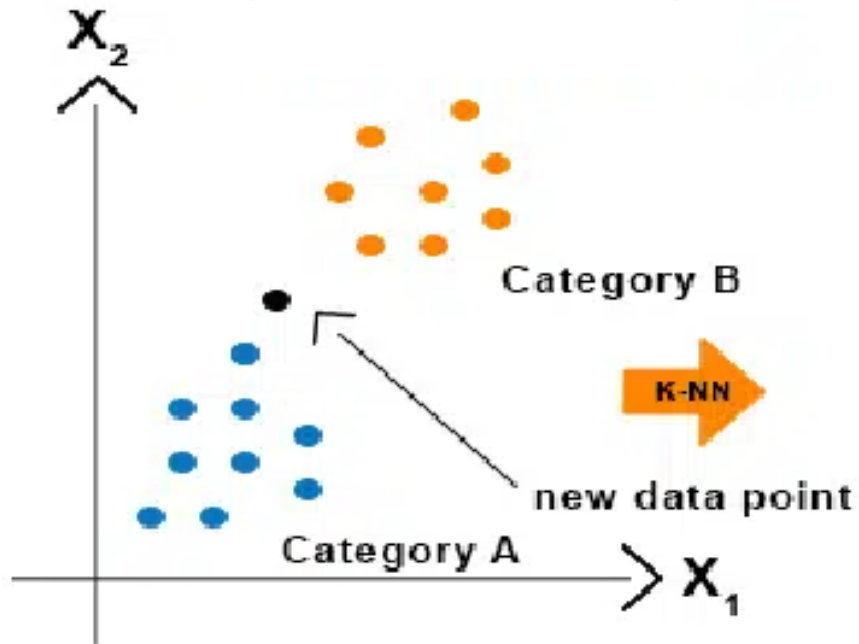
Overview

- The k-nearest neighbors (KNN) algorithm is a simple, easy-to-implement supervised machine learning algorithm that can be used to solve both classification and regression problems
- However, it is more widely used in classification problems in the industry
- It belongs to the supervised learning domain and finds intense application in pattern recognition, data mining and intrusion detection
- The KNN algorithm assumes that similar things exist in close proximity. In other words, similar things are near to each other.

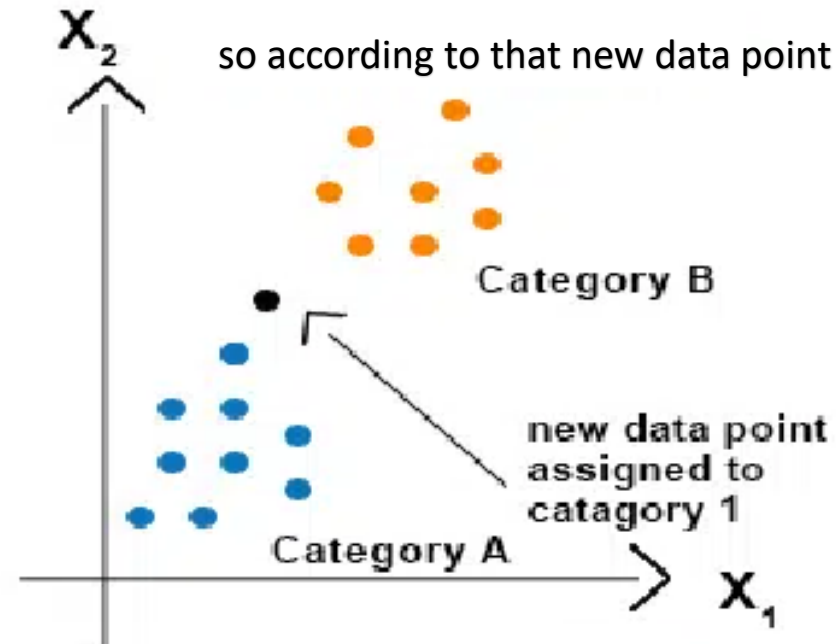


How does it work?

Initially there are two categories
New data point value is to be predicted



Distance from New data point to category A nearest value is calculated
Distance from New data point to category B nearest value is calculated



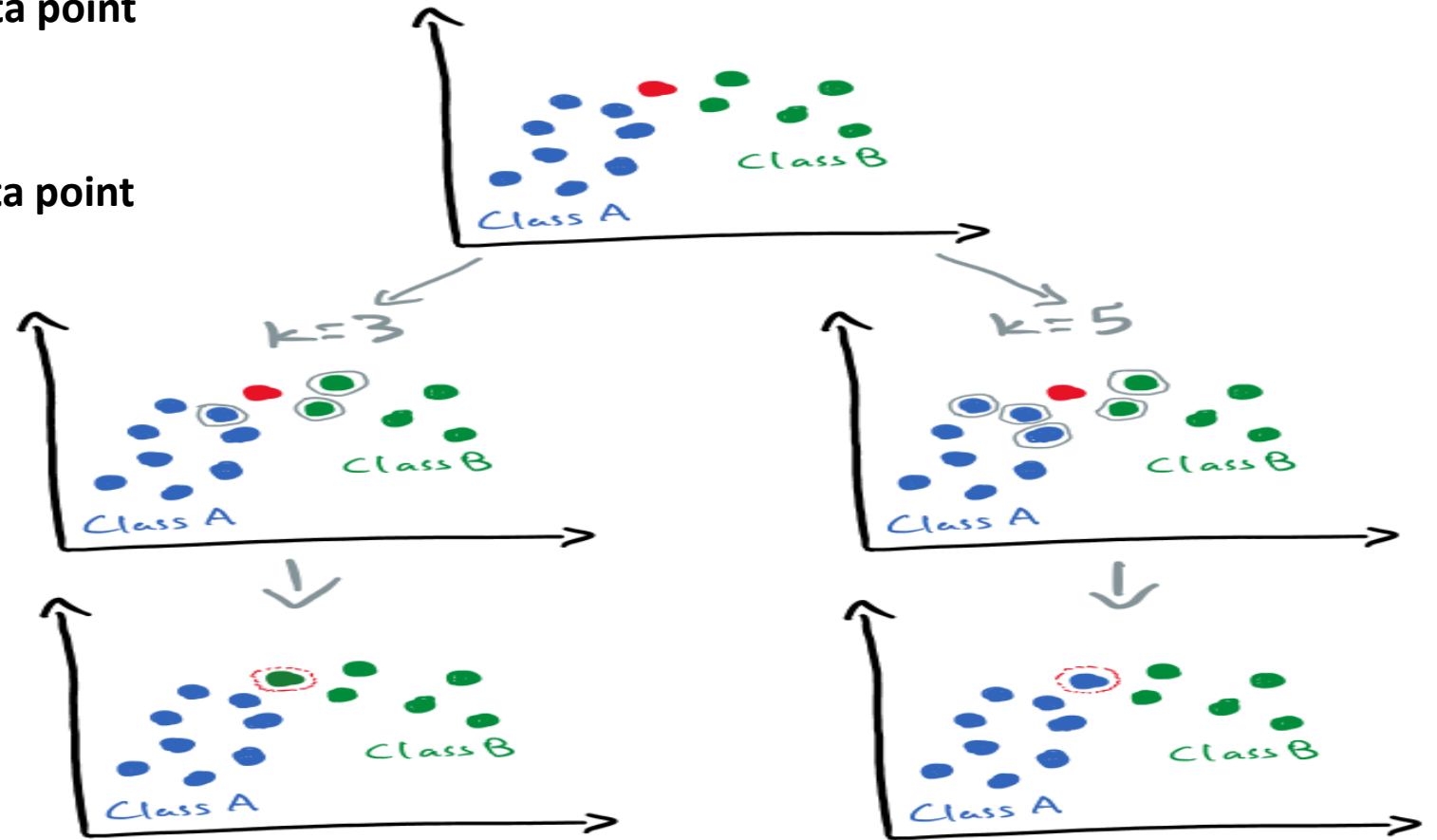
How does it work?

$K = 3$

We have to find the three closest data points
(three nearest neighbors) to the new (red) data point

$K = 5$

We have to find the five closest data points
(three nearest neighbors) to the new (red) data point



How does it work ?

- A case is classified by a majority vote of its neighbours, with the case being assigned to the class most common amongst its K nearest neighbours measured by a distance function
- If $K = 1$, then the case is simply assigned to the class of its nearest neighbour.
- Note: all three distance measures are only valid for continuous variables.

Distance functions

Euclidean	$\sqrt{\sum_{i=1}^k (x_i - y_i)^2}$
Manhattan	$\sum_{i=1}^k x_i - y_i $
Minkowski	$\left(\sum_{i=1}^k (x_i - y_i)^q \right)^{1/q}$



How does it work ?

- Choosing the optimal value for K is best done by first inspecting the data
- In general, a large K value is more precise as it reduces the overall noise but there is no guarantee
- Cross-validation is another way to retrospectively determine a good K value by using an independent dataset to validate the K value
- Historically, the optimal K for most datasets has been between 3-10. That produces much better results than 1NN.



Applications of KNN

- Recommender system
- Relevant document classification
- OCR

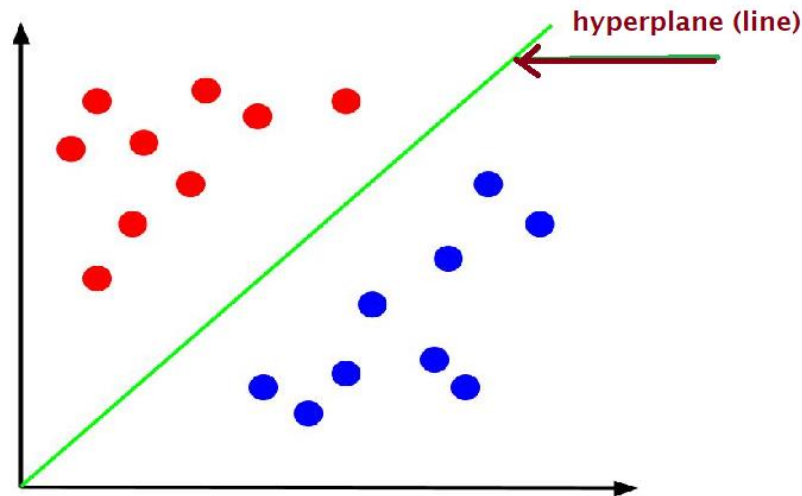


Support Vector Machine (SVM)



Overview

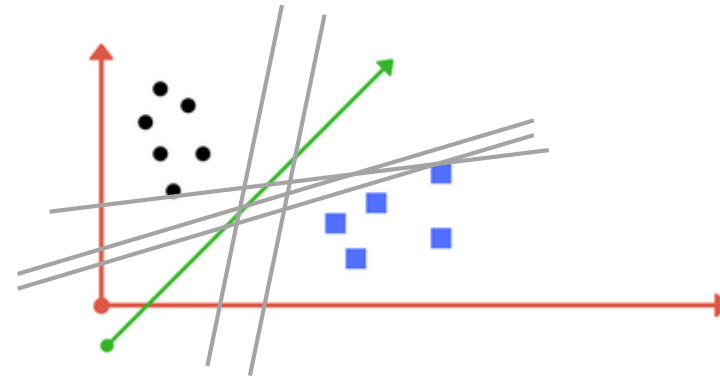
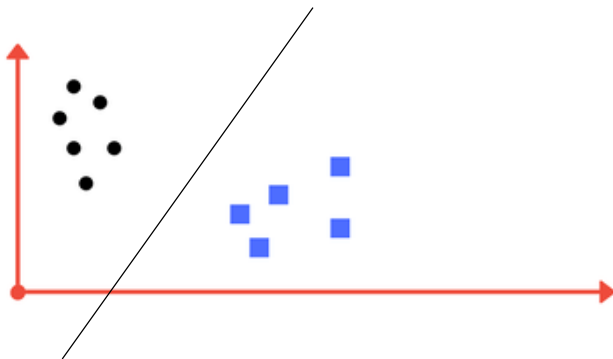
- It is a supervised machine learning algorithm that can be used for both classification and regression
- However, it is mostly used in **classification problems**
- The objective of the support vector machine algorithm is to find a **hyperplane** in an N-dimensional space (N — the number of features) that distinctly classifies the data points



How does it work ?

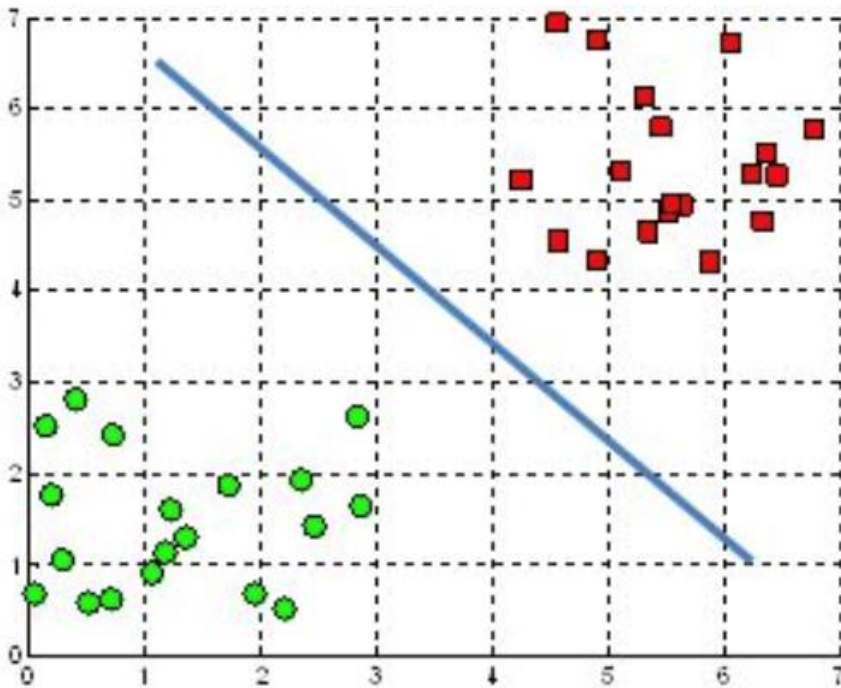
- SVM separates the classes using hyperplane
- In two dimensional space this hyperplane is a line dividing a plane in two parts where in each class lay in either side
- To separate the two classes of data points, there are many possible hyperplanes that could be chosen
- Our objective is to find a plane that has the maximum margin, i.e the maximum distance between data points of both classes

If hyperplane is chosen in this way

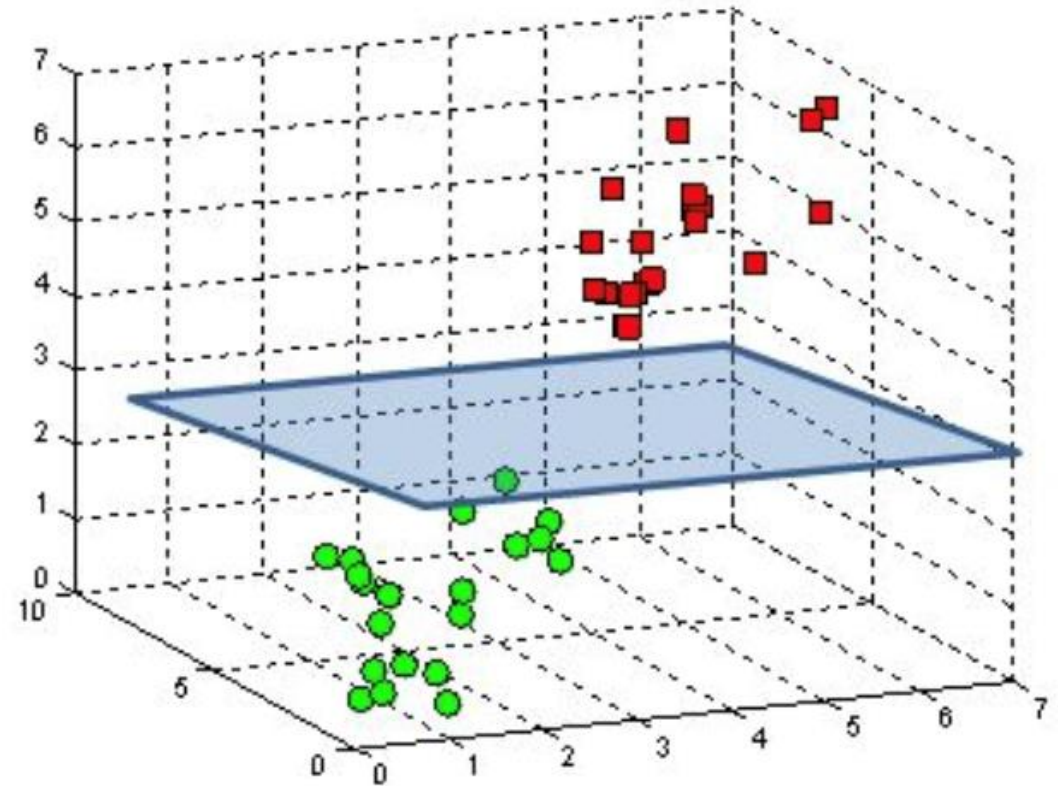


Hyperplane

A hyperplane in $\mathbb{R}^2$ is a line

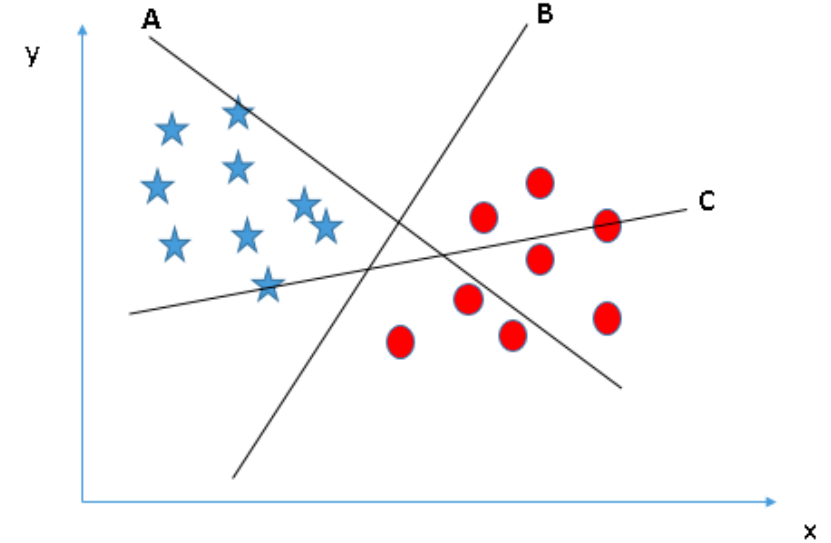


A hyperplane in $\mathbb{R}^3$ is a plane



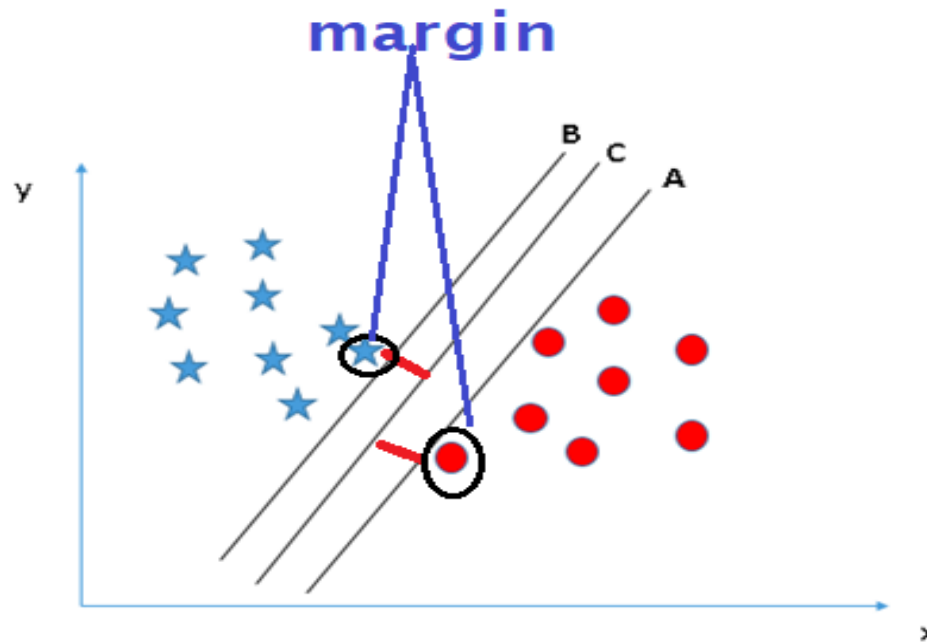
Scenario 1

- Here, we have three hyper-planes (A, B and C)
- Now, identify the right hyper-plane to classify star and circle
- You need to remember a thumb rule to identify the right hyper-plane
 - **Select the hyper-plane which segregates the two classes better (No mis-classification)**
- In this scenario, **hyper-plane “B”** has excellently performed this job



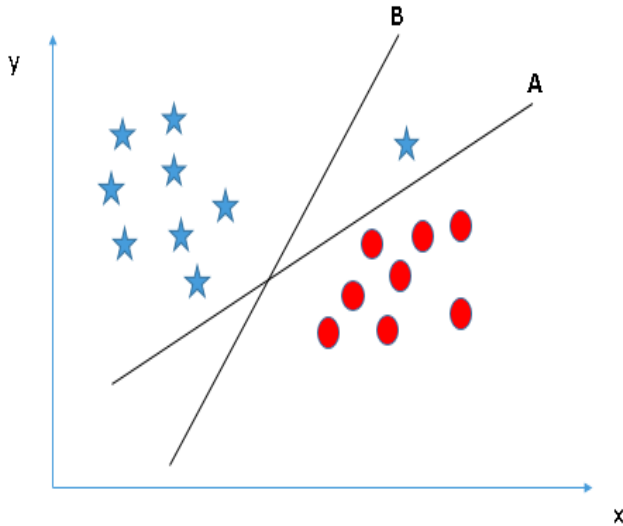
Scenario 2

- Here, we have three hyper-planes (A, B and C) and all are segregating the classes well
- Now, How can we identify the right hyper-plane?
- Here, maximizing the distances between nearest data point (either class) and hyper-plane will help us to decide the right hyper-plane. This distance is called as **Margin**.

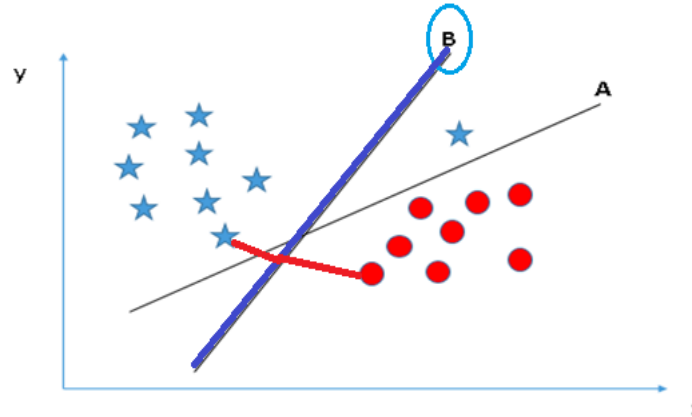


Scenario 3

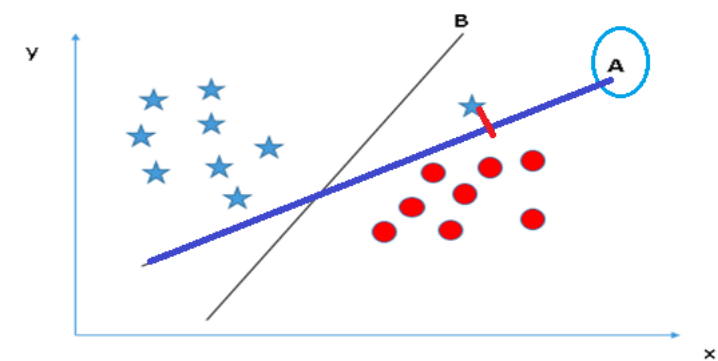
- Use the rules as discussed in previous section to identify the right hyper-plane
- Some of you may have selected the hyper-plane **B** as it has higher margin compared to **A**.
- But, SVM selects the hyper-plane which classifies the classes accurately prior to maximizing margin
- Here, hyper-plane B has a classification error and A has classified all correctly
- Therefore, the right hyper-plane is **A**.



if, B is selected as hyperplane

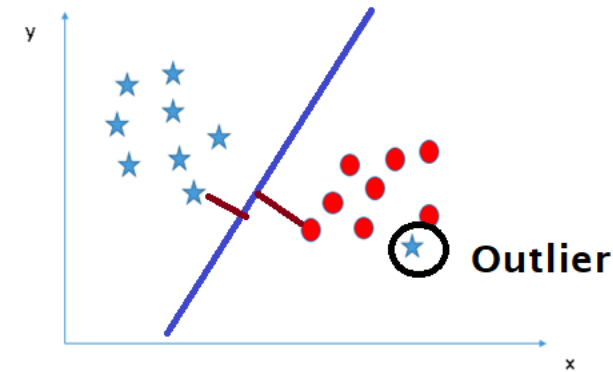
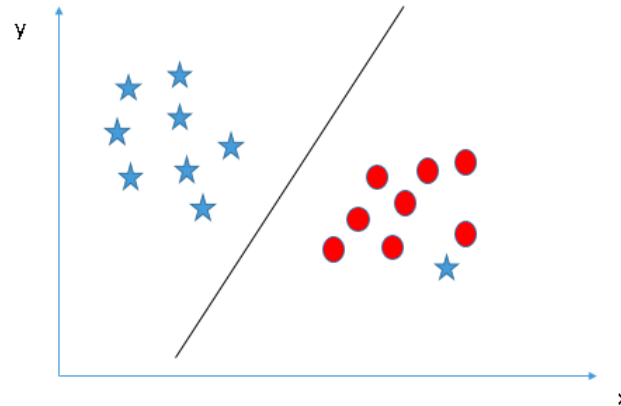


if, A is selected as hyperplane



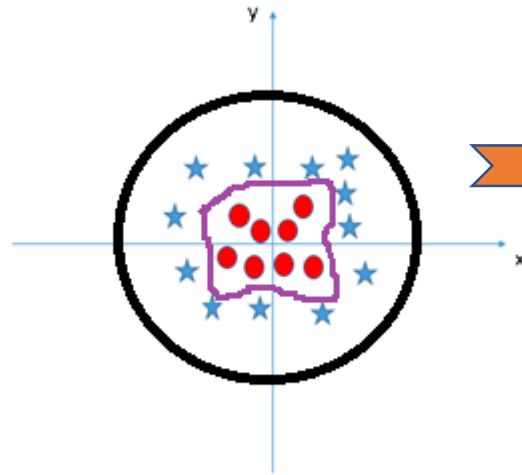
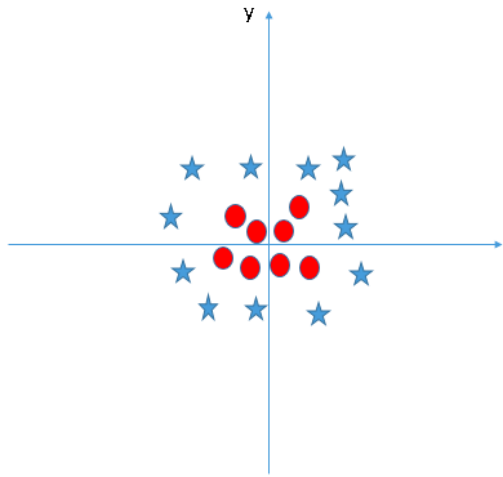
Scenario 4

- Unable to segregate the two classes using a straight line, as one of star lies in the territory of other(circle) class as an outlier
- SVM has a feature to ignore outliers and find the hyper-plane that has maximum margin
- Hence, we can say, SVM is robust to **outliers**.

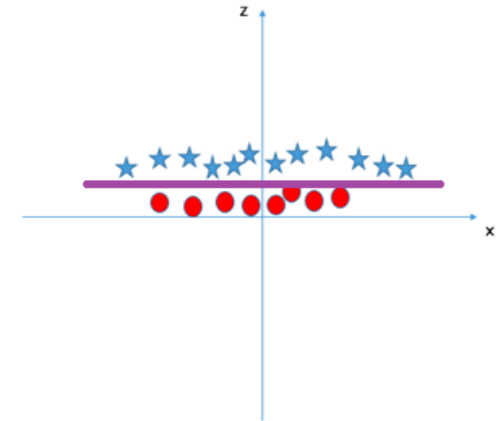
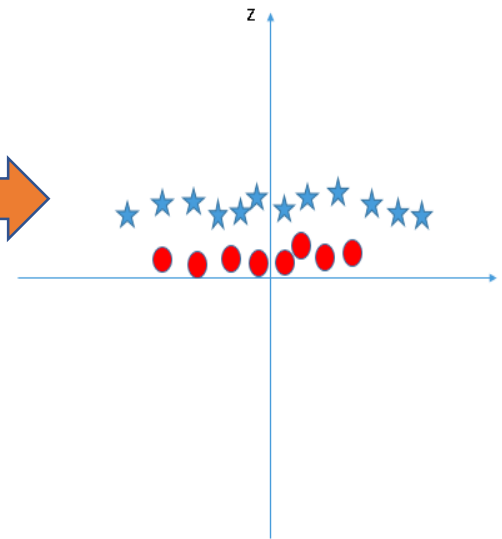


Scenario 5

- In the scenario below, we can't have linear hyper-plane between the two classes

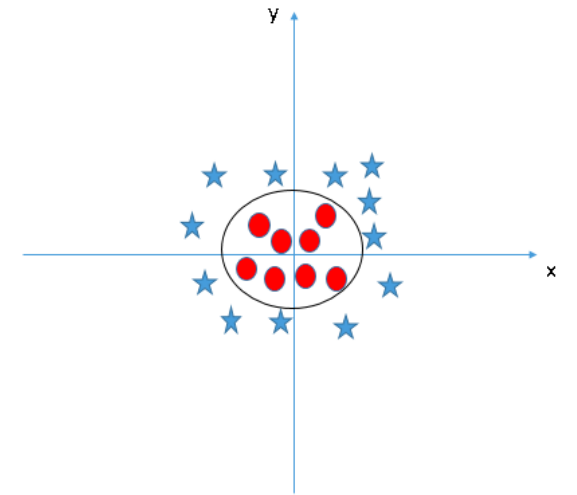


- SVM solves this problem by introducing **kernel**
 $z = x^2 + y^2$



Scenario 5

- Points to consider are:
 - All values for z would be positive always because z is the squared sum of both x and y
 - In the original plot, red circles appear close to the origin of x and y axes, leading to lower value of z and star relatively away from the origin result to higher value of z .
- In SVM, it is easy to have a linear hyper-plane between these two classes, but for such scenarios, SVM uses a trick called as **Kernel**.
- These are functions which takes low dimensional input space and transform it to a higher dimensional space, i.e. it converts non separable problem to separable problem
- It is mostly useful in non-linear separation problem
- Simply put, it does some extremely complex data transformations, then find out the process to separate the data based on the labels or outputs you've defined.



Tuning Parameters - Kernels

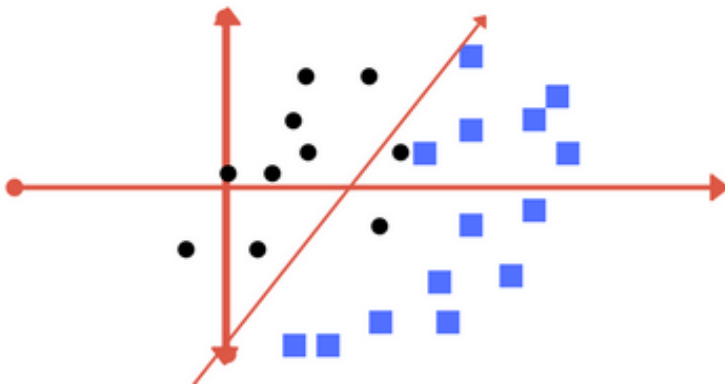
- The learning of the hyperplane in linear SVM is done by transforming the problem using some linear algebra. This is where the kernel plays role.



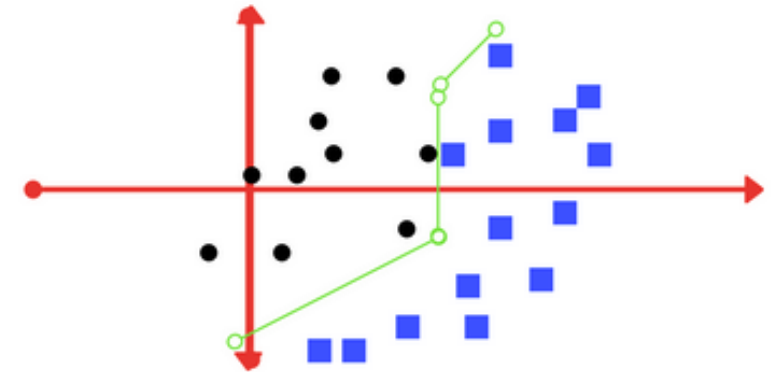
Tuning Parameters - Regularization

- The Regularization parameter tells the SVM optimization how much you want to avoid misclassifying each training example
- For large values of C , the optimization will choose a smaller-margin hyperplane if that hyperplane does a better job of getting all the training points classified correctly
- Conversely, a very small value of C will cause the optimizer to look for a larger-margin separating hyperplane, even if that hyperplane misclassifies more points

Low regularization

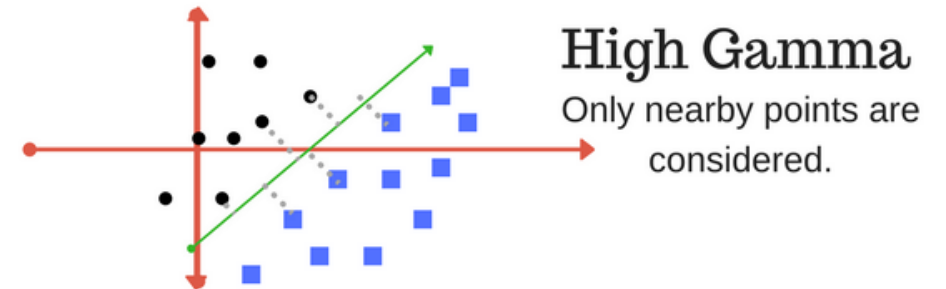
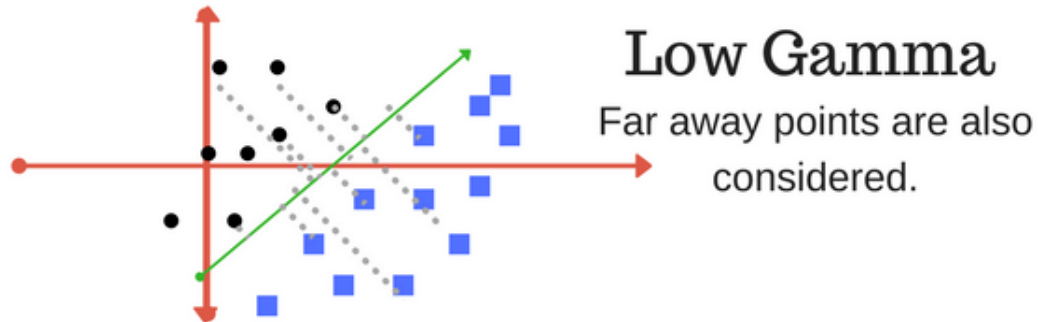


High regularization



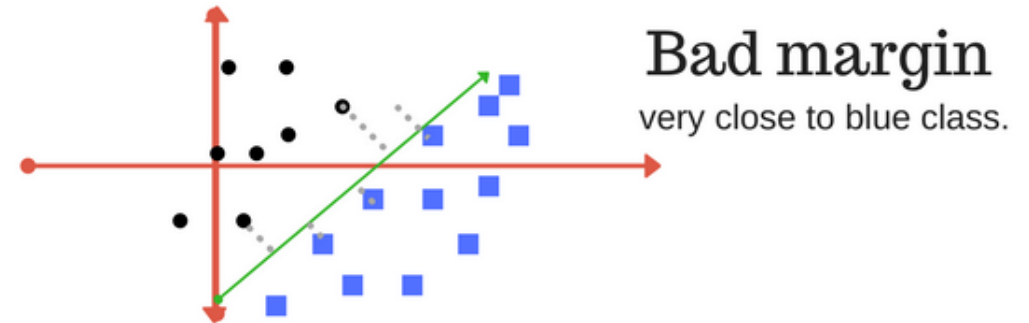
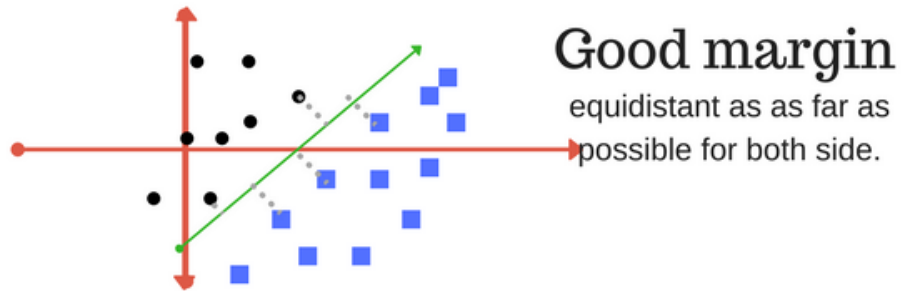
Tuning Parameters - Gamma

- The gamma parameter defines how far the influence of a single training example reaches, with low values meaning 'far' and high values meaning 'close'
- In other words
 - With low gamma, points far away from plausible separation line are also considered in calculation for the separation line
 - Where as high gamma means the points close to plausible line are considered in calculation.



Tuning Parameters - Margin

- A margin is a separation of line to the closest class points
- A good margin is one where this separation is larger for both the classes



Ensemble Learning



Overview

- Ensemble is the art of combining diverse set of learners (individual models like decision tree, logistic regression, knn) to improvise on the stability and predictive power of the model.
- The ensemble methods in machine learning combine the insights obtained from multiple learning models to facilitate accurate and improved decisions.
- Primarily used to improve the (classification, prediction, function approximation, etc.) performance of a model, or reduce the likelihood of an unfortunate selection of a poor one
- The ensemble methods in machine learning combine the insights obtained from multiple learning models to facilitate accurate and improved decisions.
- In learning models, noise, variance, and bias are the major sources of error. The ensemble methods in machine learning help minimize these error-causing factors, thereby ensuring the accuracy and stability of machine learning (ML) algorithms.
- Other applications of ensemble learning include assigning a confidence to the decision made by the model, selecting optimal (or near optimal) features, data fusion, incremental learning, nonstationary learning and error-correcting



Basic Ensemble Methods

Mode

- "mode" is the number or value that most often appears in a dataset of numbers or values.
- In this ensemble technique, machine learning professionals use a number of models for making predictions about each data point.
- The predictions made by different models are taken as separate votes.
- Subsequently, the prediction made by most models is treated as the ultimate prediction.

Mean/Average

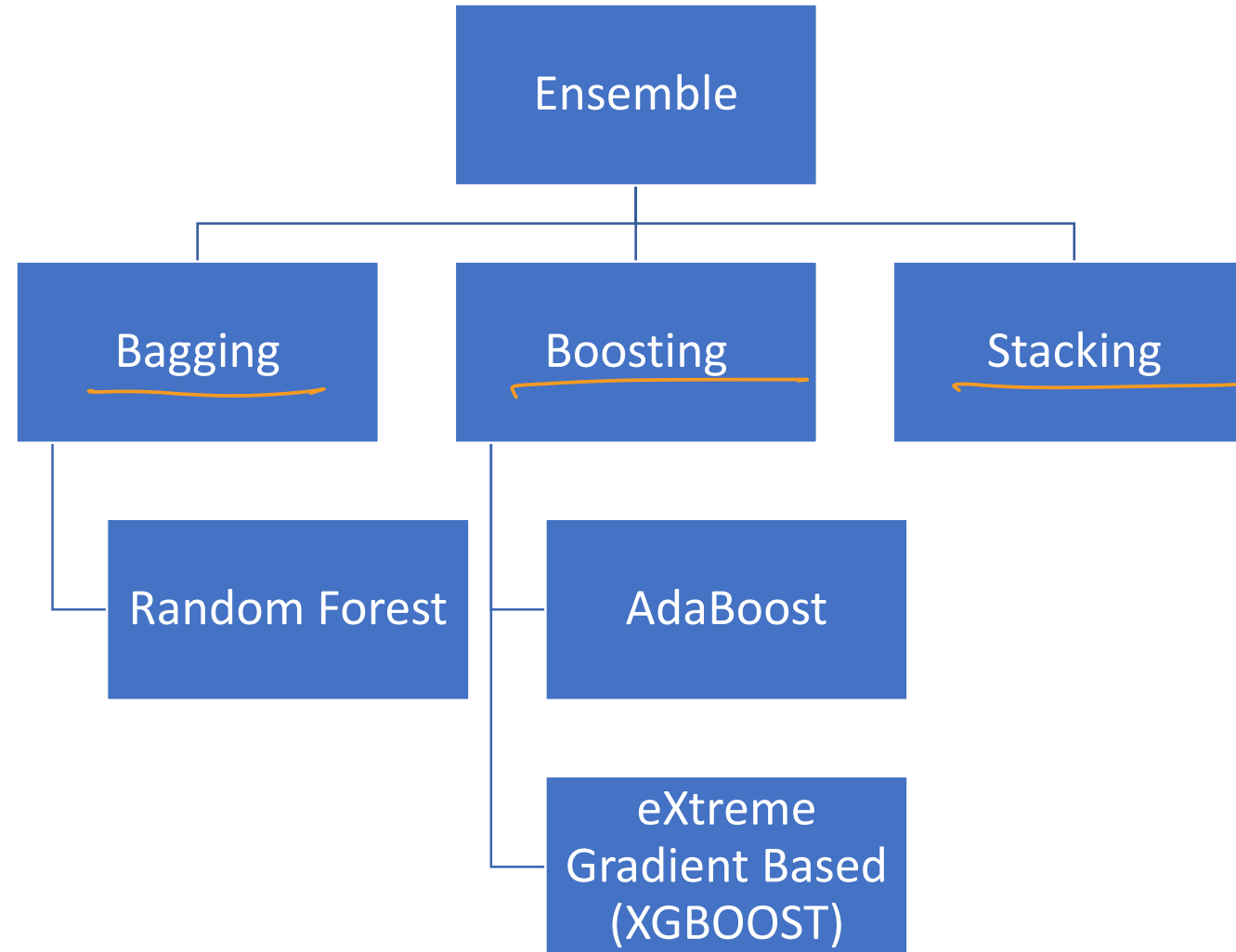
- ensemble technique, data analysts take the average predictions made by all models into account when making the ultimate prediction.

The Weighted Average

- data scientists assign different weights to all the models in order to make a prediction, where the assigned weight defines the relevance of each model.



Advance Ensemble Learning Methods



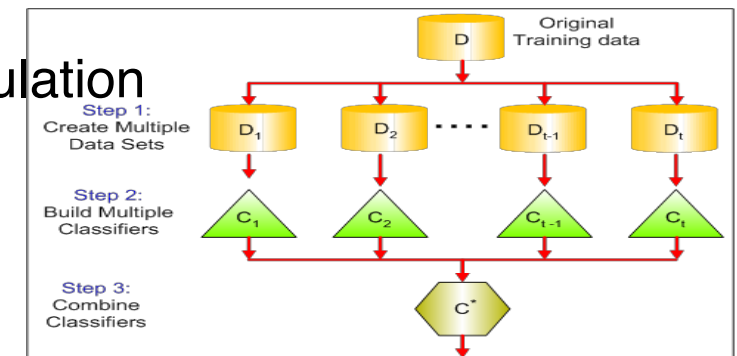
Three main classes of Ensemble Learning

- Bagging((Bootstrap Aggregating) involves fitting many decision trees on different samples of the same dataset and averaging the predictions.
- Stacking involves fitting many different models types on the same data and using another model to learn how to best combine the predictions. An iterative ensemble technique, "boosting," adjusts an observation's weight based on its last classification.
- Boosting involves adding ensemble members sequentially that correct the predictions made by prior models and outputs a weighted average of the predictions.



Bagging

- Bagging tries to implement similar learners/ same algorithm on small sample populations and then takes a mean of all the predictions
- The primary goal of "bagging" or "bootstrap aggregating" ensemble method is to minimize variance errors in decision trees.
- The objective here is to randomly create samples of training datasets with replacement (subsets of the training data).
- The subsets are then used for training decision trees or models.
- Consequently, there is a combination of multiple models, which reduces variance, as the average prediction generated from different models is much more reliable and robust than a single model or a decision tree.
- In generalized bagging, you can use different learners on different population
- This helps us to reduce the variance error
- Algorithm
 - Random Forest (decision tree)



Boosting

- Boosting refers to a family of algorithms that are able to convert weak learners to strong learners
- Boosting is an iterative technique which adjust the weight of an observation based on the last classification
- An iterative ensemble technique, "boosting," adjusts an observation's weight based on its last classification.
- In case observation is incorrectly classified, "boosting" increases the observation's weight, and vice versa.
- Boosting algorithms reduce bias errors and produce strong predictive models.
- If an observation was classified incorrectly, it tries to increase the weight of this observation and vice versa
- Algorithms
 - AdaBoost
 - Gradient Boosting
 - eXtreme Gradient Boosting

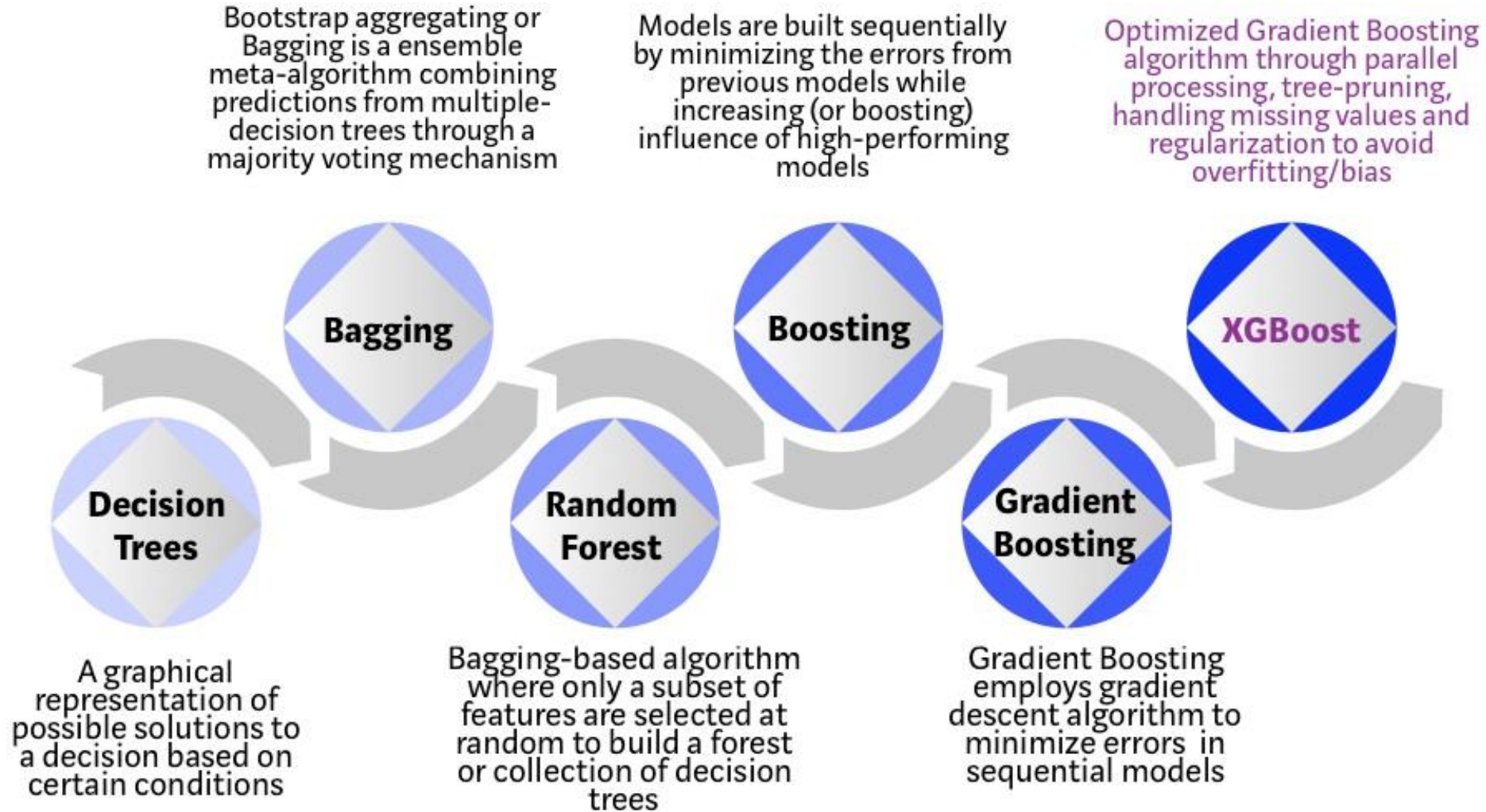


Overview

- XGBoost is a decision-tree-based ensemble Machine Learning algorithm that uses a gradient boosting framework
- XGBoost algorithm was developed as a research project at the University of Washington
- Since its introduction, this algorithm has not only been credited with winning numerous Kaggle competitions but also for being the driving force under the hood for several cutting-edge industry applications
- As a result, there is a strong community of data scientists contributing to the XGBoost open source projects with ~350 contributors and ~3,600 commits on GitHub



Evolution



Stacking

- Stacking is an ensemble learning technique that combines multiple classification or regression models via a meta-classifier or a meta-regressor
- The base level models are trained based on a complete training set, then the meta-model is trained on the outputs of the base level model as features
- The base level often consists of different learning algorithms and therefore stacking ensembles are often heterogeneous





Thank you!!!

Devendra Dhande

devendra.dhande@sunbeaminfo.com